Beyond Small Strain: Proving the importance of Finite Strain Theory
PGE 383 - Advanced Geomechanics Course Project

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Abstract

Introduction: Classical one-dimensional consolidation theory is commonly formulated using small-strain kinematics, where the soil geometry, drainage path length, and material configuration are assumed to remain essentially unchanged during deformation. This assumption is appropriate when displacements and strains are small, but it becomes restrictive for soft, highly compressible soils or heavily loaded deposits. In such cases, the distinction between the reference configuration and the deformed configuration becomes important, and geometric nonlinearity must be included through finite-strain kinematics. This work examines the limitations of the small-strain consolidation formulation and demonstrates why finite-strain theory is required when large settlement, evolving pore structure, and changing drainage geometry influence the coupled hydromechanical response.

Methods: A coupled hydromechanical finite element formulation is developed and compared for both small-strain and large-strain consolidation. The small-strain model is written in terms of the infinitesimal Cauchy strain tensor and the total Cauchy stress, while the large-strain model uses the deformation gradient, Jacobian, Green–Lagrange strain, and Piola–Kirchhoff stress measures to track deformation relative to the reference configuration. The governing equilibrium and mass-conservation equations are converted from strong form to weak form using test functions, surface integrals, and virtual strain measures. Numerical simulations are performed for homogeneous and bilayer one-dimensional soil columns under different surcharge loads. The homogeneous cases are compared with the classical Terzaghi analytical solution, while the bilayer cases are used to investigate the role of stratigraphy, stiffness contrast, void ratio, and permeability contrast between a soft clay layer and a stiff clay foundation.

Results: For small applied loads, the small-strain and large-strain solutions show excellent agreement, confirming that the small-strain formulation is a reliable approximation when deformation remains negligible. However, as the applied load increases, the two formulations begin to diverge. In the classical homogeneous model, the small-strain solution increasingly overpredicts the consolidation response relative to the large-strain solution because it neglects geometric shortening and nonlinear strain effects. In the bilayer model, the difference is more pronounced because the highly compressible soft clay layer undergoes larger deformation than the stiff underlying layer. The large-strain formulation captures the evolving geometry, nonlinear strain contribution, and deformation-dependent response, whereas the small-strain formulation treats the domain and drainage path as fixed throughout the simulation.

Conclusion: The results show that small-strain Terzaghi-type consolidation theory is an effective and computationally efficient baseline for low-load conditions, stiff soils, or problems involving negligible deformation. However, for soft soils, high surcharge loading, and stratified deposits with strong material contrasts, finite-strain theory is necessary for physically consistent prediction of settlement and pore-pressure dissipation. By retaining geometric nonlinearity through the deformation gradient and Jacobian, the large-strain formulation provides a more rigorous framework for consolidation problems in which the evolving configuration of the soil skeleton directly influences the hydromechanical response.

1 Purpose and Scope

This document summarizes the continuum-mechanics background, weak forms, analytical one-dimensional Terzaghi consolidation solution, and numerical model setups used for coupled poromechanical consolidation problems. The presentation is organized so that the small-strain and large-strain formulations can be compared directly.

The main unknown fields are the solid displacement field $\mathbf{u}$ and the excess pore pressure field p. In a two-dimensional computational setting,

$$\mathbf{u} = u_x \mathbf{e}_x + u_y \mathbf{e}_y, \tag{1}$$

while one-dimensional consolidation is recovered by applying roller side constraints and appropriate hydraulic boundary conditions.

The sign convention used here is that tensile stress is positive and pore pressure is compressive. Under this convention, the total stress is decomposed as

$$\boldsymbol{\sigma}_{\text{total}} = \boldsymbol{\sigma}' - \alpha p \mathbf{I}, \tag{2}$$

where $\boldsymbol{\sigma}'$ is the effective stress, α is the Biot coefficient, p is the pore pressure, and $\mathbf{I}$ is the identity tensor.

2 Notation and Configurations

A material point is labeled by its reference coordinate $\mathbf{X} \in \Omega_0$. After deformation, the same material point occupies the spatial position $\mathbf{x} \in \Omega_t$. The motion map is

$$\boldsymbol{\varphi} : \Omega_0 \to \Omega_t, \qquad \mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t). \tag{3}$$

The reference configuration Ω_0 is the undeformed domain, and the current or deformed configuration Ω_t is the domain occupied by the solid skeleton at time t.

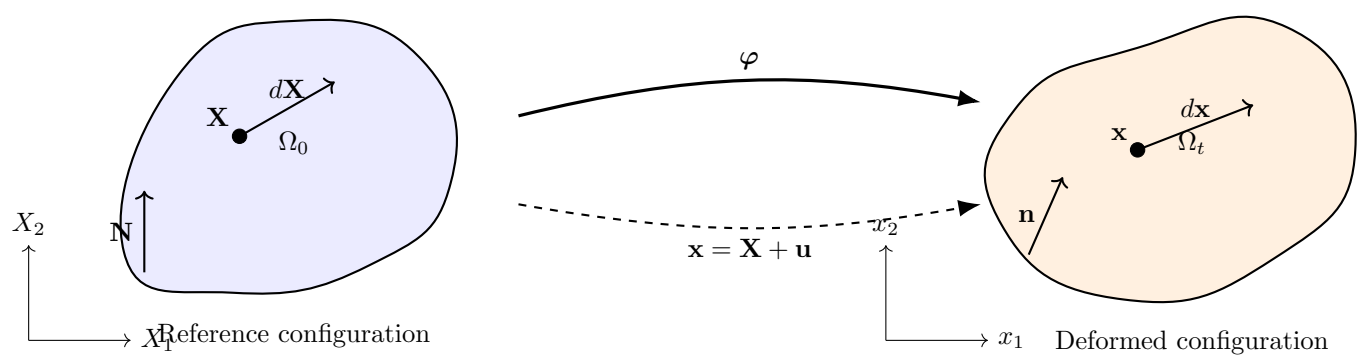

Figure 1: Reference and deformed configurations. The motion map $\boldsymbol{\varphi}$ sends each material point $\mathbf{X}$ in Ω_0 to its current position $\mathbf{x}$ in Ω_t.

The deformation gradient is the gradient of the motion with respect to the reference coordinates:

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \nabla_0 \boldsymbol{\varphi} = \mathbf{I} + \nabla_0 \mathbf{u}. \tag{4}$$

It maps differential material line elements from the reference configuration to the current configuration:

$$d\mathbf{x} = \mathbf{F}\, d\mathbf{X}. \tag{5}$$

Therefore, $\mathbf{F}$ contains local stretch, shear, and rotation information. It is central in large-strain formulations because gradients, stresses, fluxes, and volume changes must be consistently mapped between Ω_0 and Ω_t.

The Jacobian is the determinant of the deformation gradient:

$$J = \det \mathbf{F}. \tag{6}$$

It measures the local volume change:

$$dV = J\, dV_0. \tag{7}$$

Thus, $J > 1$ indicates local expansion, $J < 1$ indicates local compression, and $J = 1$ indicates locally volume-preserving deformation. In finite deformation poromechanics, J appears in the mass balance because pore volume and Darcy flow are affected by changes in the solid skeleton volume.

Several related strain measures are used below. The right Cauchy–Green tensor is

$$\mathbf{C} = \mathbf{F}^T\mathbf{F}, \tag{8}$$

and the Green–Lagrange strain tensor is

$$\mathbf{E} = \frac{1}{2}\left(\mathbf{C} - \mathbf{I}\right) = \frac{1}{2}\left(\nabla_0\mathbf{u} + \nabla_0\mathbf{u}^T + \nabla_0\mathbf{u}^T\nabla_0\mathbf{u}\right). \tag{9}$$

For small strains, the quadratic term is neglected and the infinitesimal strain tensor becomes

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right). \tag{10}$$

3 Weak Forms, Test Functions, and Boundary Integrals

A weak form is obtained by multiplying the governing differential equation by an admissible weighting function, also called a test function, and integrating over the domain. In finite element methods, the test functions are usually chosen from the same function space as the trial solution, except that they vanish on essential or Dirichlet boundaries.

For mechanics, the test function is a virtual displacement $\delta\mathbf{u}$. It satisfies

$$\delta\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_u, \tag{11}$$

because the displacement is already prescribed on Γ_u. Physically, $\delta\mathbf{u}$ represents an arbitrary kinematically admissible virtual perturbation of the displacement field.

For pore pressure, the test function is a virtual pressure field ψ. It satisfies

$$\psi = 0 \quad \text{on } \Gamma_p, \tag{12}$$

where pore pressure is prescribed. The weak form must hold for every admissible $\delta\mathbf{u}$ and every admissible ψ.

Surface integrals arise naturally when integration by parts is applied to divergence terms. For example,

$$\int_\Omega (\nabla\cdot\boldsymbol{\sigma})\cdot\delta\mathbf{u}\, dV = \int_{\partial\Omega} (\boldsymbol{\sigma}\mathbf{n})\cdot\delta\mathbf{u}\, dA - \int_\Omega \boldsymbol{\sigma} : \nabla\delta\mathbf{u}\, dV. \tag{13}$$

The boundary term contains the traction vector

$$\mathbf{t} = \boldsymbol{\sigma}\mathbf{n}, \tag{14}$$

where $\mathbf{n}$ is the outward unit normal on the current boundary. Similarly, flux boundary terms contain $\mathbf{q} \cdot \mathbf{n}$ or, in the reference configuration, $\mathbf{Q} \cdot \mathbf{N}$.

Virtual test strains are the strain measures associated with the virtual displacement field. In the small-strain setting,

$$\delta\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\delta\mathbf{u} + \nabla\delta\mathbf{u}^T\right). \tag{15}$$

In the large-strain Total Lagrangian setting,

$$\delta\mathbf{F} = \nabla_0\delta\mathbf{u}, \qquad \delta\mathbf{E} = \frac{1}{2}\left(\mathbf{F}^T\delta\mathbf{F} + \delta\mathbf{F}^T\mathbf{F}\right). \tag{16}$$

These virtual strains are not independent physical strains; they are the variations of the strain measures induced by a virtual displacement. They appear in the internal virtual work terms.

4 Stress Measures in Small and Large Strain

4.1 Cauchy Stress

The Cauchy stress $\boldsymbol{\sigma}$ is the physical stress acting on surfaces in the current configuration. It relates the current traction vector to the current outward unit normal:

$$\mathbf{t} = \boldsymbol{\sigma}\mathbf{n}. \tag{17}$$

Cauchy stress is therefore a spatial stress measure. It is natural in small-strain formulations and in Eulerian descriptions.

4.2 First Piola–Kirchhoff Stress

The first Piola–Kirchhoff stress, denoted $\mathbf{P}$, maps reference area normals to current force vectors. If $\bar{\mathbf{t}}_0$ is the nominal traction per unit reference area, then

$$\bar{\mathbf{t}}_0 = \mathbf{P}\mathbf{N}, \tag{18}$$

where $\mathbf{N}$ is the outward unit normal in the reference configuration. The PK1 stress is related to the Cauchy stress by

$$\mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-T}. \tag{19}$$

The tensor $\mathbf{P}$ is generally not symmetric because it connects quantities living in two different configurations: forces in the current configuration and areas in the reference configuration. It is work-conjugate to the deformation gradient:

$$\mathbf{P} : \delta\mathbf{F}. \tag{20}$$

4.3 Second Piola–Kirchhoff Stress

The second Piola–Kirchhoff stress, denoted $\mathbf{S}$, is a fully material stress measure. It is related to the PK1 stress through

$$\mathbf{P} = \mathbf{F}\mathbf{S}, \tag{21}$$

or equivalently,

$$\mathbf{S} = \mathbf{F}^{-1}\mathbf{P} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-T}. \tag{22}$$

For hyperelastic or elastic materials without couple stresses, $\mathbf{S}$ is symmetric and is work-conjugate to the Green–Lagrange strain:

$$\mathbf{S} : \delta\mathbf{E}. \tag{23}$$

This makes $\mathbf{S}$ convenient in Total Lagrangian formulations, where all volume integrals are evaluated over the reference configuration.

5 Small-Strain Poromechanics Weak Form

5.1 Solid Mechanics: Equilibrium

The static equilibrium equation for the porous medium is given by the strong form

$$\nabla \cdot \boldsymbol{\sigma}_{\text{total}} + \mathbf{b} = \mathbf{0} \qquad \text{in } \Omega. \tag{24}$$

Here, $\boldsymbol{\sigma}_{\text{total}}$ is the total Cauchy stress tensor and $\mathbf{b}$ is the body force vector per unit current volume.

Let $\delta\mathbf{u}$ be a kinematically admissible virtual displacement field that vanishes on the Dirichlet boundary Γ_u. Multiplying the strong form by $\delta\mathbf{u}$ and integrating over the domain Ω gives

$$\int_\Omega (\nabla \cdot \boldsymbol{\sigma}_{\text{total}}) \cdot \delta\mathbf{u}\, dV + \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, dV = 0. \tag{25}$$

Applying the product rule and the divergence theorem to the stress term gives

$$\int_{\partial\Omega} (\boldsymbol{\sigma}_{\text{total}}\mathbf{n}) \cdot \delta\mathbf{u}\, dA - \int_\Omega \boldsymbol{\sigma}_{\text{total}} : \nabla(\delta\mathbf{u})\, dV + \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, dV = 0. \tag{26}$$

Because $\boldsymbol{\sigma}_{\text{total}}$ is symmetric, its double contraction with $\nabla(\delta\mathbf{u})$ equals its double contraction with the symmetric virtual strain tensor:

$$\delta\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla\delta\mathbf{u} + \nabla\delta\mathbf{u}^T\right). \tag{27}$$

Applying the Neumann boundary condition

$$\bar{\mathbf{t}} = \boldsymbol{\sigma}_{\text{total}}\mathbf{n} \qquad \text{on } \Gamma_t, \tag{28}$$

the weak form becomes

$$-\int_\Omega \boldsymbol{\sigma}_{\text{total}} : \delta\boldsymbol{\varepsilon}\, dV + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \delta\mathbf{u}\, dA + \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, dV = 0. \tag{29}$$

Using the effective stress relation

$$\boldsymbol{\sigma}_{\text{total}} = \boldsymbol{\sigma}' - \alpha p\mathbf{I}, \tag{30}$$

and the linear elastic effective stress law

$$\boldsymbol{\sigma}' = \mathbb{C} : \boldsymbol{\varepsilon}, \tag{31}$$

we obtain

$$-\int_\Omega (\boldsymbol{\sigma}' - \alpha p \mathbf{I}) : \delta\boldsymbol{\varepsilon}\, dV + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \delta\mathbf{u}\, dA + \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, dV = 0. \tag{32}$$

Expanding the stress term and using

$$\mathbf{I} : \delta\boldsymbol{\varepsilon} = \mathrm{tr}(\delta\boldsymbol{\varepsilon}) = \nabla \cdot \delta\mathbf{u}, \tag{33}$$

we get

$$-\int_\Omega \boldsymbol{\sigma}' : \delta\boldsymbol{\varepsilon}\, dV + \int_\Omega \alpha p (\nabla \cdot \delta\mathbf{u})\, dV + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \delta\mathbf{u}\, dA + \int_\Omega \mathbf{b} \cdot \delta\mathbf{u}\, dV = 0. \tag{34}$$

Rearranging gives the small-strain mechanics weak form:

$$\boxed{\int_\Omega \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma}'\, dV - \int_\Omega \alpha p (\nabla \cdot \delta\mathbf{u})\, dV = \int_\Omega \delta\mathbf{u} \cdot \mathbf{b}\, dV + \int_{\Gamma_t} \delta\mathbf{u} \cdot \bar{\mathbf{t}}\, dA.} \tag{35}$$

5.2 Fluid Flow: Mass Conservation

The conservation of fluid mass in a deforming porous medium, assuming small strain, is expressed as

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \mathbf{q} = 0, \tag{36}$$

where ζ is the variation of fluid content and $\mathbf{q}$ is the Darcy flux vector.

The constitutive relationships for fluid content and Darcy flow are

$$\zeta = \alpha \varepsilon_v + S_\epsilon p = \alpha(\nabla \cdot \mathbf{u}) + S_\epsilon p, \tag{37}$$

$$\mathbf{q} = -\frac{\mathbf{k}}{\mu_f} \nabla p. \tag{38}$$

Here S_ϵ is the constrained specific storage, $\mathbf{k}$ is the intrinsic permeability tensor, and μ_f is the fluid viscosity. The subscript on μ_f is used to distinguish fluid viscosity from the shear modulus μ used later in the elastic material model.

Substituting these relations into the mass conservation equation gives the strong form

$$\alpha \nabla \cdot \dot{\mathbf{u}} + S_\epsilon \dot{p} - \nabla \cdot \left(\frac{\mathbf{k}}{\mu_f} \nabla p \right) = 0. \tag{39}$$

Let ψ be an admissible pore-pressure test function. Multiplying the strong form by ψ and integrating over Ω gives

$$\int_\Omega \psi \alpha (\nabla \cdot \dot{\mathbf{u}})\, dV + \int_\Omega \psi S_\epsilon \dot{p}\, dV - \int_\Omega \psi \nabla \cdot \left(\frac{\mathbf{k}}{\mu_f} \nabla p \right) dV = 0. \tag{40}$$

Applying integration by parts only to the flux-divergence term gives

$$-\int_\Omega \psi \nabla \cdot \mathbf{q}\, dV = \int_\Omega \nabla \psi \cdot \mathbf{q}\, dV - \int_{\partial\Omega} \psi (\mathbf{q} \cdot \mathbf{n})\, dA. \tag{41}$$

Using $\mathbf{q} = -(\mathbf{k}/\mu_f)\nabla p$ and defining the prescribed inward normal flux on Γ_q as

$$\bar{q}_n = -\mathbf{q} \cdot \mathbf{n}, \tag{42}$$

we obtain the small-strain pressure weak form:

$$\boxed{\int_\Omega \psi\alpha(\nabla \cdot \dot{\mathbf{u}})\, dV + \int_\Omega \psi S_\epsilon \dot{p}\, dV + \int_\Omega \nabla\psi \cdot \left(\frac{\mathbf{k}}{\mu_f}\nabla p\right) dV = \int_{\Gamma_q} \psi\bar{q}_n\, dA.} \tag{43}$$

6 Large-Strain Total Lagrangian Poromechanics Weak Form

6.1 Solid Mechanics: Equilibrium

In a Total Lagrangian formulation, equilibrium is written over the reference configuration Ω_0:

$$\nabla_0 \cdot \mathbf{P} + \mathbf{b}_0 = \mathbf{0} \qquad \text{in } \Omega_0, \tag{44}$$

where $\nabla_0\cdot$ is the divergence operator with respect to $\mathbf{X}$, $\mathbf{P}$ is the first Piola–Kirchhoff stress tensor, and $\mathbf{b}_0$ is the body force per unit reference volume.

Let $\delta\mathbf{u}$ be a kinematically admissible virtual displacement field with $\delta\mathbf{u} = \mathbf{0}$ on $\Gamma_{u,0}$. Multiplying by $\delta\mathbf{u}$ and integrating over Ω_0 gives

$$\int_{\Omega_0} (\nabla_0 \cdot \mathbf{P}) \cdot \delta\mathbf{u}\, dV_0 + \int_{\Omega_0} \mathbf{b}_0 \cdot \delta\mathbf{u}\, dV_0 = 0. \tag{45}$$

Integration by parts gives

$$\int_{\partial\Omega_0} (\mathbf{PN}) \cdot \delta\mathbf{u}\, dA_0 - \int_{\Omega_0} \mathbf{P} : \nabla_0(\delta\mathbf{u})\, dV_0 + \int_{\Omega_0} \mathbf{b}_0 \cdot \delta\mathbf{u}\, dV_0 = 0. \tag{46}$$

Using

$$\bar{\mathbf{t}}_0 = \mathbf{PN} \qquad \text{on } \Gamma_{t,0}, \tag{47}$$

and

$$\delta\mathbf{F} = \nabla_0\delta\mathbf{u}, \tag{48}$$

we obtain

$$-\int_{\Omega_0} \mathbf{P} : \delta\mathbf{F}\, dV_0 + \int_{\Gamma_{t,0}} \bar{\mathbf{t}}_0 \cdot \delta\mathbf{u}\, dA_0 + \int_{\Omega_0} \mathbf{b}_0 \cdot \delta\mathbf{u}\, dV_0 = 0. \tag{49}$$

The PK1 stress is computationally inconvenient because it is generally asymmetric. Using

$$\mathbf{P} = \mathbf{F}\mathbf{S}_{\text{total}}, \tag{50}$$

the internal work term becomes

$$\mathbf{P} : \delta\mathbf{F} = (\mathbf{F}\mathbf{S}_{\text{total}}) : \delta\mathbf{F} = \mathbf{S}_{\text{total}} : (\mathbf{F}^T\delta\mathbf{F}). \tag{51}$$

Because $\mathbf{S}_{\text{total}}$ is symmetric, only the symmetric part of $\mathbf{F}^T\delta\mathbf{F}$ contributes. This symmetric part is the virtual Green–Lagrange strain:

$$\delta\mathbf{E} = \frac{1}{2}\left(\mathbf{F}^T\delta\mathbf{F} + \delta\mathbf{F}^T\mathbf{F}\right). \tag{52}$$

Therefore,

$$\mathbf{P} : \delta\mathbf{F} = \mathbf{S}_{\text{total}} : \delta\mathbf{E}. \tag{53}$$

The weak form can now be written as

$$\int_{\Omega_0} \delta\mathbf{E} : \mathbf{S}_{\text{total}}\, dV_0 = \int_{\Omega_0} \delta\mathbf{u} \cdot \mathbf{b}_0\, dV_0 + \int_{\Gamma_{t,0}} \delta\mathbf{u} \cdot \bar{\mathbf{t}}_0\, dA_0. \tag{54}$$

To couple fluid pressure, the spatial effective stress principle is pulled back to the reference configuration. Starting from

$$\boldsymbol{\sigma}_{\text{total}} = \boldsymbol{\sigma}' - \alpha p \mathbf{I}, \tag{55}$$

use the Piola pull-back

$$\mathbf{S} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-T}. \tag{56}$$

This gives

$$\begin{aligned}
\mathbf{S}_{\text{total}} &= J\mathbf{F}^{-1}\boldsymbol{\sigma}_{\text{total}}\mathbf{F}^{-T} && (57)\\
&= J\mathbf{F}^{-1}\left(\boldsymbol{\sigma}' - \alpha p\mathbf{I}\right)\mathbf{F}^{-T} && (58)\\
&= \mathbf{S}' - \alpha p J\mathbf{F}^{-1}\mathbf{F}^{-T} && (59)\\
&= \mathbf{S}' - \alpha p J\mathbf{C}^{-1}. && (60)
\end{aligned}$$

Here $\mathbf{S}'$ is the effective PK2 stress. For a linear elastic material written in Green–Lagrange strain,

$$\mathbf{S}' = \mathbb{C} : \mathbf{E}. \tag{61}$$

Substituting the pressure split into the internal work gives the large-strain mechanics weak form:

$$\boxed{\int_{\Omega_0} \delta\mathbf{E} : \mathbf{S}'\, dV_0 - \int_{\Omega_0} \alpha p J\, \delta\mathbf{E} : \mathbf{C}^{-1}\, dV_0 = \int_{\Omega_0} \delta\mathbf{u} \cdot \mathbf{b}_0\, dV_0 + \int_{\Gamma_{t,0}} \delta\mathbf{u} \cdot \bar{\mathbf{t}}_0\, dA_0.} \tag{62}$$

Equivalently, the compact total-stress form is

$$\boxed{\int_{\Omega_0} \delta\mathbf{E} : \mathbf{S}_{\text{total}}\, dV_0 = \int_{\Omega_0} \delta\mathbf{u} \cdot \mathbf{b}_0\, dV_0 + \int_{\Gamma_{t,0}} \delta\mathbf{u} \cdot \bar{\mathbf{t}}_0\, dA_0.} \tag{63}$$

6.2 Fluid Flow: Mass Conservation

The strong form of fluid mass conservation in the current configuration Ω_t is

$$\alpha(\nabla_x \cdot \mathbf{v}_s) + \nabla_x \cdot \mathbf{q} + S_\epsilon \dot{p} = 0, \tag{64}$$

where $\mathbf{v}_s = \dot{\mathbf{u}}$ is the solid skeleton velocity, $\mathbf{q}$ is the spatial Darcy flux, and S_ϵ is the constrained specific storage.

To map this equation to the Total Lagrangian configuration, multiply by J:

$$\alpha J(\nabla_x \cdot \mathbf{v}_s) + J\nabla_x \cdot \mathbf{q} + JS_\epsilon \dot{p} = 0. \tag{65}$$

Using Euler's identity for volumetric change,

$$\dot{J} = J\nabla_x \cdot \mathbf{v}_s, \tag{66}$$

and the Piola identity for flux divergence,

$$J\nabla_x \cdot \mathbf{q} = \nabla_0 \cdot \mathbf{Q}, \qquad \mathbf{Q} = J\mathbf{F}^{-1}\mathbf{q}, \tag{67}$$

the Total Lagrangian strong form is

$$\alpha \dot{J} + \nabla_0 \cdot \mathbf{Q} + JS_\epsilon \dot{p} = 0. \tag{68}$$

The spatial Darcy law is

$$\mathbf{q} = -\frac{\mathbf{k}}{\mu_f}\nabla_x p. \tag{69}$$

The pressure gradient transforms as

$$\nabla_x p = \mathbf{F}^{-T}\nabla_0 p. \tag{70}$$

Therefore, the nominal flux is

$$\begin{aligned} \mathbf{Q} &= J\mathbf{F}^{-1}\mathbf{q} && (71)\\ &= J\mathbf{F}^{-1}\left(-\frac{\mathbf{k}}{\mu_f}\mathbf{F}^{-T}\nabla_0 p\right) && (72)\\ &= -\frac{\mathbf{K}_{\mathrm{TL}}}{\mu_f}\nabla_0 p, && (73) \end{aligned}$$

where the pulled-back permeability tensor is

$$\mathbf{K}_{\mathrm{TL}} = J\mathbf{F}^{-1}\mathbf{k}\mathbf{F}^{-T}. \tag{74}$$

Let ψ be an admissible pore-pressure test function. Multiplying the Total Lagrangian strong form by ψ and integrating over Ω_0 gives

$$\int_{\Omega_0} \psi\alpha\dot{J}\, dV_0 + \int_{\Omega_0} \psi\nabla_0 \cdot \mathbf{Q}\, dV_0 + \int_{\Omega_0} \psi JS_\epsilon\dot{p}\, dV_0 = 0. \tag{75}$$

Equivalently, substituting $\mathbf{Q} = -(\mathbf{K}_{\mathrm{TL}}/\mu_f)\nabla_0 p$ gives

$$\int_{\Omega_0} \psi\alpha\dot{J}\, dV_0 - \int_{\Omega_0} \psi\nabla_0 \cdot \left(\frac{\mathbf{K}_{\mathrm{TL}}}{\mu_f}\nabla_0 p\right) dV_0 + \int_{\Omega_0} \psi JS_\epsilon\dot{p}\, dV_0 = 0. \tag{76}$$

Applying integration by parts to the flux term gives

$$-\int_{\Omega_0} \psi\nabla_0 \cdot \mathbf{Q}\, dV_0 = \int_{\Omega_0} \nabla_0\psi \cdot \mathbf{Q}\, dV_0 - \int_{\partial\Omega_0} \psi(\mathbf{Q}\cdot\mathbf{N})\, dA_0. \tag{77}$$

Defining the prescribed inward nominal flux by

$$\bar{Q}_n = -\mathbf{Q}\cdot\mathbf{N} \qquad \text{on } \Gamma_{q,0}, \tag{78}$$

we obtain the large-strain pressure weak form:

$$\boxed{\underbrace{\int_{\Omega_0} \psi\alpha\dot{J}\, dV_0}_{\text{volumetric coupling}} + \underbrace{\int_{\Omega_0} \psi JS_\epsilon\dot{p}\, dV_0}_{\text{fluid storage}} + \underbrace{\int_{\Omega_0} \nabla_0\psi \cdot \left(\frac{\mathbf{K}_{\mathrm{TL}}}{\mu_f}\nabla_0 p\right) dV_0}_{\text{Darcy flow}} = \int_{\Gamma_{q,0}} \psi\bar{Q}_n\, dA_0.} \tag{79}$$

7 Analytical Solution of One-Dimensional Terzaghi Consolidation

The classical one-dimensional Terzaghi consolidation equation for excess pore water pressure $u(z,t)$ is the diffusion equation

$$\frac{\partial u}{\partial t} = C_v \frac{\partial^2 u}{\partial z^2}, \tag{80}$$

where C_v is the coefficient of consolidation.

For a soil layer of thickness H with a drained top surface and an impermeable bottom boundary, the boundary conditions are

$$u(0,t) = 0, \qquad \text{drained top surface}, \tag{81}$$

$$\frac{\partial u}{\partial z}(H,t) = 0, \qquad \text{impermeable bottom boundary}. \tag{82}$$

For an instantaneous uniform load, the initial condition is

$$u(z,0) = u_0, \qquad 0 < z \leq H. \tag{83}$$

We use separation of variables. Let

$$u(z,t) = Z(z)T(t). \tag{84}$$

Substitution into the governing equation gives

$$Z\frac{dT}{dt} = C_v T \frac{d^2 Z}{dz^2}. \tag{85}$$

Dividing by $C_v ZT$ gives

$$\frac{1}{C_v T}\frac{dT}{dt} = \frac{1}{Z}\frac{d^2 Z}{dz^2} = -\lambda^2. \tag{86}$$

The negative separation constant is chosen so that pore pressure decays with time. This yields two ordinary differential equations:

$$\frac{dT}{dt} + C_v \lambda^2 T = 0, \tag{87}$$

$$\frac{d^2 Z}{dz^2} + \lambda^2 Z = 0. \tag{88}$$

The temporal solution is

$$T(t) = \exp\bigl(-C_v \lambda^2 t\bigr), \tag{89}$$

and the spatial solution is

$$Z(z) = A\cos(\lambda z) + B\sin(\lambda z). \tag{90}$$

Applying the top boundary condition $Z(0) = 0$ gives

$$A\cos(0) + B\sin(0) = 0 \quad \Rightarrow \quad A = 0, \tag{91}$$

so

$$Z(z) = B\sin(\lambda z). \tag{92}$$

Applying the bottom boundary condition $Z'(H) = 0$ gives

$$B\lambda\cos(\lambda H) = 0. \tag{93}$$

For a nontrivial solution, $B \neq 0$ and $\lambda \neq 0$, so

$$\cos(\lambda H) = 0. \tag{94}$$

Hence,

$$\lambda_m H = \frac{\pi}{2} + m\pi = \frac{\pi}{2}(2m+1), \qquad m = 0, 1, 2, \ldots. \tag{95}$$

Define

$$M_m = \frac{\pi}{2}(2m+1), \qquad \lambda_m = \frac{M_m}{H}. \tag{96}$$

The general solution is therefore

$$u(z,t) = \sum_{m=0}^{\infty} B_m \sin\left(\frac{M_m z}{H}\right) \exp\left(-\frac{M_m^2 C_v t}{H^2}\right). \tag{97}$$

To find B_m, apply the initial condition:

$$u_0 = \sum_{m=0}^{\infty} B_m \sin\left(\frac{M_m z}{H}\right). \tag{98}$$

Multiplying both sides by $\sin(M_n z/H)$ and integrating from 0 to H gives

$$\int_0^H u_0 \sin\left(\frac{M_n z}{H}\right) dz = \sum_{m=0}^{\infty} B_m \int_0^H \sin\left(\frac{M_m z}{H}\right) \sin\left(\frac{M_n z}{H}\right) dz. \tag{99}$$

Using orthogonality, only the $m = n$ term remains:

$$\int_0^H u_0 \sin\left(\frac{M_m z}{H}\right) dz = B_m \int_0^H \sin^2\left(\frac{M_m z}{H}\right) dz. \tag{100}$$

Evaluate the two integrals:

$$\int_0^H u_0 \sin\left(\frac{M_m z}{H}\right) dz = u_0 \left[-\frac{H}{M_m} \cos\left(\frac{M_m z}{H}\right)\right]_0^H \tag{101}$$

$$= \frac{u_0 H}{M_m}, \tag{102}$$

because $\cos(M_m) = 0$ and $\cos(0) = 1$. Also,

$$\int_0^H \sin^2\left(\frac{M_m z}{H}\right) dz = \frac{H}{2}. \tag{103}$$

Thus,

$$B_m = \frac{2u_0}{M_m}. \tag{104}$$

Introducing the dimensionless time factor

$$T_v = \frac{C_v t}{H^2}, \tag{105}$$

the final analytical series solution is

$$\boxed{u(z,t) = \sum_{m=0}^{\infty} \frac{2u_0}{M_m} \sin\left(\frac{M_m z}{H}\right) \exp\left(-M_m^2 T_v\right), \qquad M_m = \frac{\pi}{2}(2m+1).} \tag{106}$$

8 Numerical Model Setups

This section documents the four consolidation configurations used for comparison. There are two homogeneous-column models and two bilayer-column models. Each material layout is studied in both small-strain and large-strain form so that the effects of stratigraphy and finite-deformation kinematics can be separated.

Table 1: Organization of the consolidation model configurations.

Case	Stratigraphy	Kinematics	Main feature
Homogeneous small strain	Single layer	Infinitesimal strain	Fixed domain and fixed h
Homogeneous large strain	Single layer	Total Lagrangian finite strain	Evolving volume ratio, vo
Bilayer small strain	Soft clay over stiff clay	Infinitesimal strain	Layer contrast on a fixed
Bilayer large strain	Soft clay over stiff clay	Total Lagrangian finite strain	Layer contrast with evolvi

In the small-strain models, the mesh is interpreted as a fixed spatial domain. The strain measure is the infinitesimal strain $\boldsymbol{\varepsilon}$, the hydraulic mobility is constant, and pore-volume change enters through $\nabla \cdot \mathbf{u}$. In the large-strain models, the calculation is written on the reference configuration Ω_0. The motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$, the deformation gradient $\mathbf{F}$, and the Jacobian $J = \det \mathbf{F}$ update the current geometry, the local pore volume, and any deformation-dependent permeability or mobility.

8.1 Homogeneous Small-Strain Consolidation Model

Geometry and spatial discretization. The homogeneous small-strain model represents a one-dimensional vertical soil column with total drainage thickness $H = 10.0\,\mathrm{m}$. The computational domain has width $1.0\,\mathrm{m}$ and is discretized using a single vertical column of 300 nine-node quadrilateral elements (QUAD9). The lateral roller constraints enforce one-dimensional compression in a two-dimensional finite element mesh.

Material properties. The domain consists of one homogeneous poroelastic material. To make this case directly comparable with the homogeneous large-strain model, the same baseline elastic and hydraulic constants are used: shear modulus $\mu = 2000\,\mathrm{kPa}$, Lamé parameter $\lambda = 0.0\,\mathrm{kPa}$, Biot coefficient $\alpha = 1.0$, and constant fluid mobility

$$\frac{\kappa}{\mu_f} = 10^{-6}\,\frac{\mathrm{m}^2}{\mathrm{kPa} \cdot \mathrm{day}}. \tag{107}$$

Here μ_f denotes the fluid viscosity, while μ denotes the solid shear modulus. In the small-strain setting, the mobility and domain size remain fixed throughout the simulation; there is no update through J.

Initial and boundary conditions. The initial state has zero displacement and zero excess pore pressure. The side boundaries are rollers with $u_x = 0$, the bottom boundary is mechanically rigid and hydraulically impermeable, and the top boundary is drained with $p = 0$. The schematic shows a uniform compressive surcharge $q = -500\,\mathrm{kPa}$ so that it can be compared directly with the homogeneous large-strain model.

Purpose of this case. This model is the closest numerical counterpart to classical Terzaghi consolidation. Because the geometry, permeability, and storage parameters are fixed, the pressure dissipation is governed by the small-strain weak form and a constant consolidation process. It is therefore useful as a baseline before adding either finite-deformation effects or material layering.

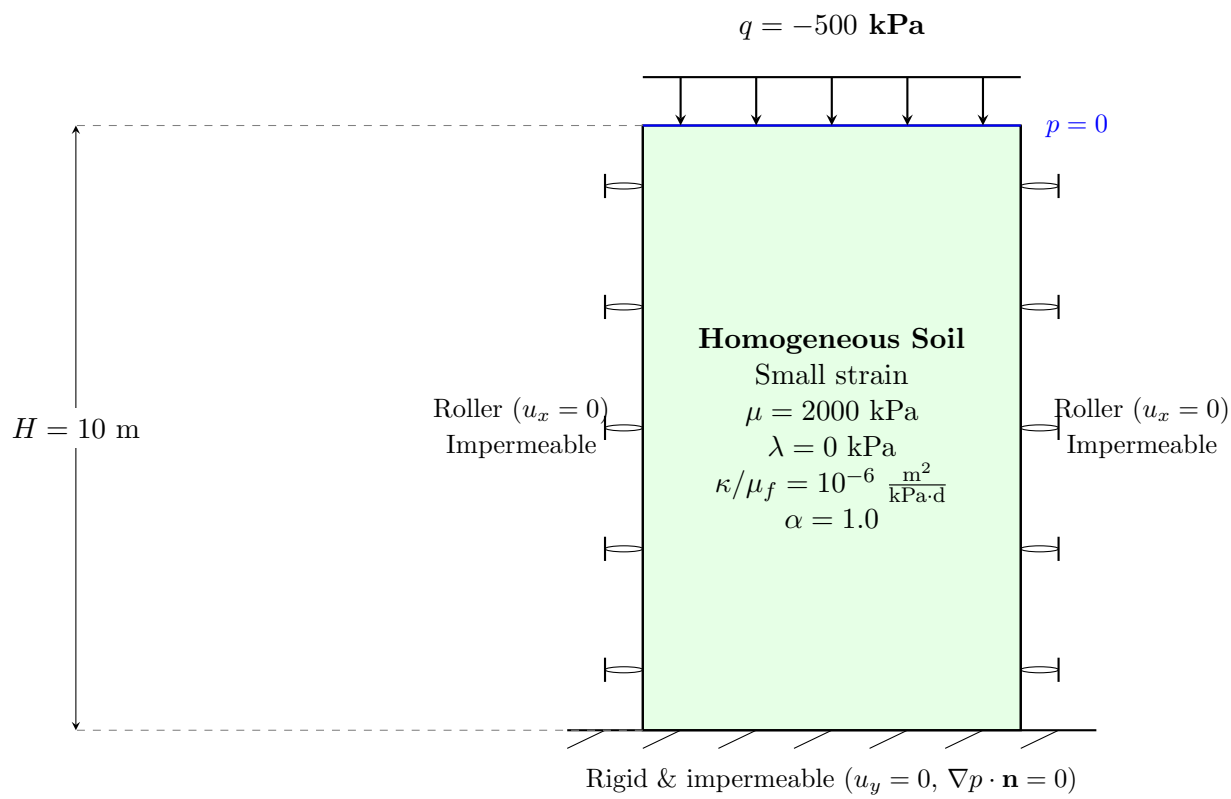

Figure 2: Schematic representation of the one-dimensional homogeneous small-strain consolidation model.

8.2 Small-Strain Bilayer Consolidation Model

Geometry and spatial discretization. The small-strain model represents a one-dimensional vertical soil column with total thickness $H = 5.0\,\mathrm{m}$. The computational domain has width $1.0\,\mathrm{m}$ and is discretized using a single vertical column of 300 nine-node quadrilateral elements (QUAD9).

Stratigraphy and material properties. The domain is divided into a two-layer stratified system:

- **Layer 1: Soft clay, top.** This layer occupies $z = 3.0\,\mathrm{m}$ to $z = 5.0\,\mathrm{m}$. It uses linear elasticity with shear modulus $\mu = 362.25\,\mathrm{kPa}$, Lamé parameter $\lambda = 0.0\,\mathrm{kPa}$, and constant fluid mobility $\kappa/\mu_f = 10^{-6}\,\mathrm{m}^2/(\mathrm{kPa} \cdot \mathrm{day})$.
- **Layer 2: Stiff clay, bottom.** This layer occupies $z = 0.0\,\mathrm{m}$ to $z = 3.0\,\mathrm{m}$. It uses shear modulus $\mu = 1293.75\,\mathrm{kPa}$, Lamé parameter $\lambda = 0.0\,\mathrm{kPa}$, and lower constant fluid mobility $\kappa/\mu_f = 10^{-7}\,\mathrm{m}^2/(\mathrm{kPa} \cdot \mathrm{day})$.

In the small-strain framework, material properties and the computational domain remain fixed throughout the simulation. The Biot coefficient is $\alpha = 1.0$. Fluid storage is fully accommodated by volumetric deformation of the skeleton, with numerical specific storage $m_v = 0.0$.

Initial and boundary conditions. The system is initialized at rest with zero displacement and zero excess pore water pressure. The lateral mechanical boundaries are rollers, so $u_x = 0$ while vertical settlement is allowed. The base is fixed. Hydraulically, the top boundary is drained, $p = 0$, while the side and bottom boundaries are impermeable. A vertical compressive surcharge of 100 kPa is applied on the top surface and ramped over the first 0.01 day for solver stability.

Numerical strategy. The coupled system is solved using a Preconditioned Jacobian-Free Newton–Krylov (PJFNK) method with the MUMPS parallel direct solver. Time integration is performed using implicit Euler over a 3250-day duration. An iteration-adaptive time stepper captures the early pore-pressure response, starting from $\Delta t = 0.001$ day and increasing as consolidation proceeds.

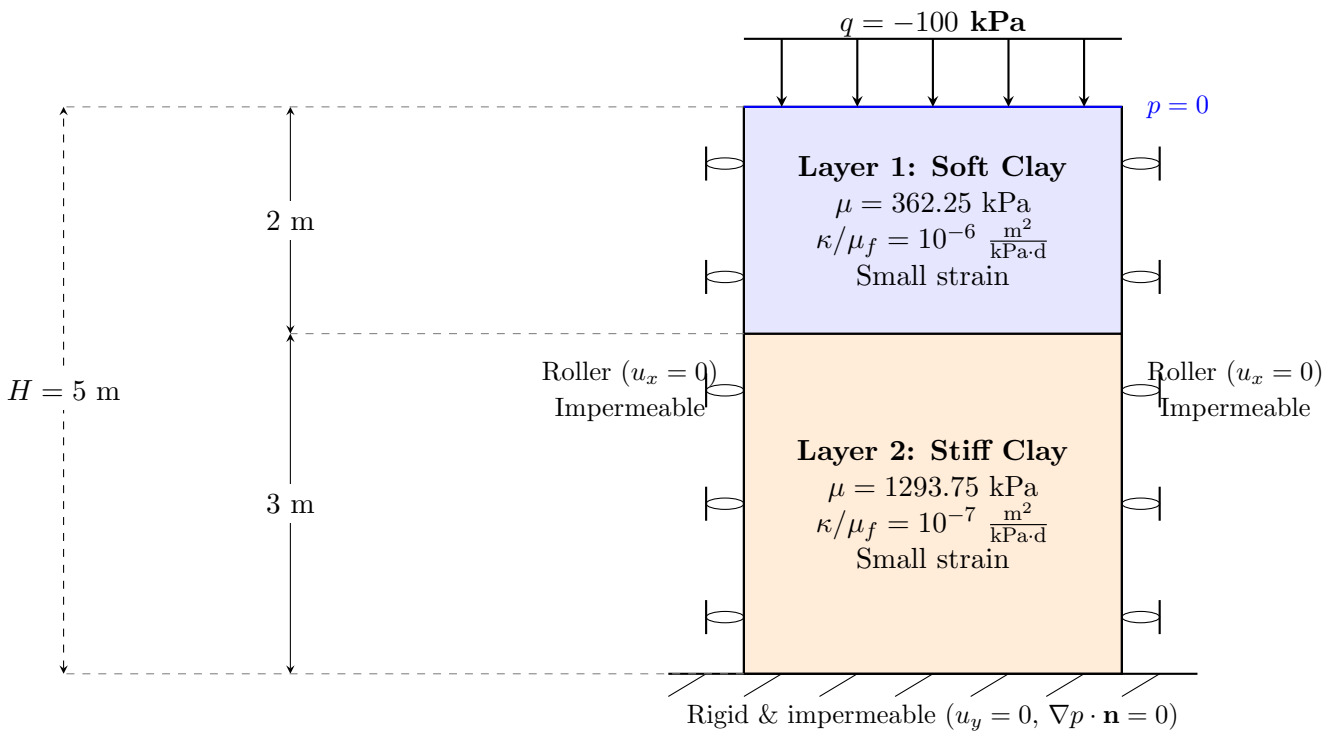

Figure 3: Small-strain bilayer consolidation model.

8.3 Large-Strain Bilayer Consolidation Model

Geometry and spatial discretization. The large-strain bilayer model uses the same initial column geometry as the small-strain bilayer model: total reference height $H = 5.0$ m, width 1.0 m, and a single vertical column of 300 QUAD9 elements. The layer interface is located at $Z = 3.0$ m in the reference configuration. During deformation, the current thickness of each layer can change because material points move according to the motion map $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$.

Stratigraphy and initial material properties. The material layout is defined in the reference configuration:

- **Layer 1: Soft clay, top.** This layer occupies $Z = 3.0\,\mathrm{m}$ to $Z = 5.0\,\mathrm{m}$. Its initial properties are shear modulus $\mu = 362.25\,\mathrm{kPa}$, initial void ratio $e_0 = 2.5$, and initial mobility $\kappa_0/\mu_f = 10^{-6}\,\mathrm{m}^2/(\mathrm{kPa}\cdot\mathrm{day})$.
- **Layer 2: Stiff clay, bottom.** This layer occupies $Z = 0.0\,\mathrm{m}$ to $Z = 3.0\,\mathrm{m}$. Its initial properties are shear modulus $\mu = 1293.75\,\mathrm{kPa}$, initial void ratio $e_0 = 0.5$, and initial mobility $\kappa_0/\mu_f = 10^{-7}\,\mathrm{m}^2/(\mathrm{kPa}\cdot\mathrm{day})$.

The subscript zero emphasizes that e_0 and κ_0/μ_f are reference-state quantities. In the large-strain formulation, the Jacobian J updates the current volume of the skeleton and pore space. Consequently, void ratio is not simply a fixed input after deformation begins. If the constitutive model links permeability to void ratio, the hydraulic mobility evolves as consolidation proceeds.

Initial and boundary conditions. The initial displacement and excess pore pressure are zero. The lateral boundaries are rollers, the bottom boundary is mechanically rigid and hydraulically impermeable, and the top boundary is drained with $p = 0$. A uniform compressive surcharge $q = -100\,\mathrm{kPa}$ is applied on the top surface. Because the formulation is Total Lagrangian, the impermeability condition is naturally written on the reference boundary as $\nabla_0 p \cdot \mathbf{N} = 0$.

Interpretation of the response. This case combines stiffness contrast, mobility contrast, and finite-deformation kinematics. The soft upper clay is more compressible and more permeable, so it tends to deform and drain more quickly. The stiff lower clay has lower initial mobility and smaller initial void ratio, so it can delay pressure dissipation and create stronger hydraulic gradients near the layer interface. Compared with the small-strain bilayer model, the large-strain bilayer model also accounts for changing layer thicknesses, changing pore volume, and possible permeability evolution.

8.4 Large-Strain Homogeneous Consolidation Model

Geometry and spatial discretization. The large-strain model represents a one-dimensional vertical soil column with total drainage thickness $H = 10.0\,\mathrm{m}$. The domain has width $1.0\,\mathrm{m}$ and is discretized using a single vertical column of 300 QUAD9 elements.

Material properties. The domain is a homogeneous poroelastic medium governed by Total Lagrangian kinematics. The solid skeleton has shear modulus $\mu = 2000\,\mathrm{kPa}$ and Lamé parameter $\lambda = 0.0\,\mathrm{kPa}$, giving an initial one-dimensional constrained modulus of $M = 4000\,\mathrm{kPa}$. The initial void ratio is $e_0 = 0.8$ and the initial intrinsic fluid mobility is $\kappa_0/\mu_f = 10^{-6}\,\mathrm{m}^2/(\mathrm{kPa}\cdot\mathrm{day})$. A small numerical volumetric compressibility $m_v = 10^{-7}\,\mathrm{kPa}^{-1}$ stabilizes the initial undrained response. The Biot coefficient is $\alpha = 1.0$. During finite deformation, the void ratio and permeability evolve dynamically.

Initial and boundary conditions. The initial state has zero displacement and zero excess pore pressure. Lateral boundaries are rollers with $u_x = 0$, the bottom boundary is fixed, and the top surface receives a uniform compressive surcharge of $500\,\mathrm{kPa}$ ramped over the first $1.0\,\mathrm{day}$. Hydraulically, the top surface is drained with $p = 0$, while the lateral and bottom boundaries are impermeable in the reference configuration.

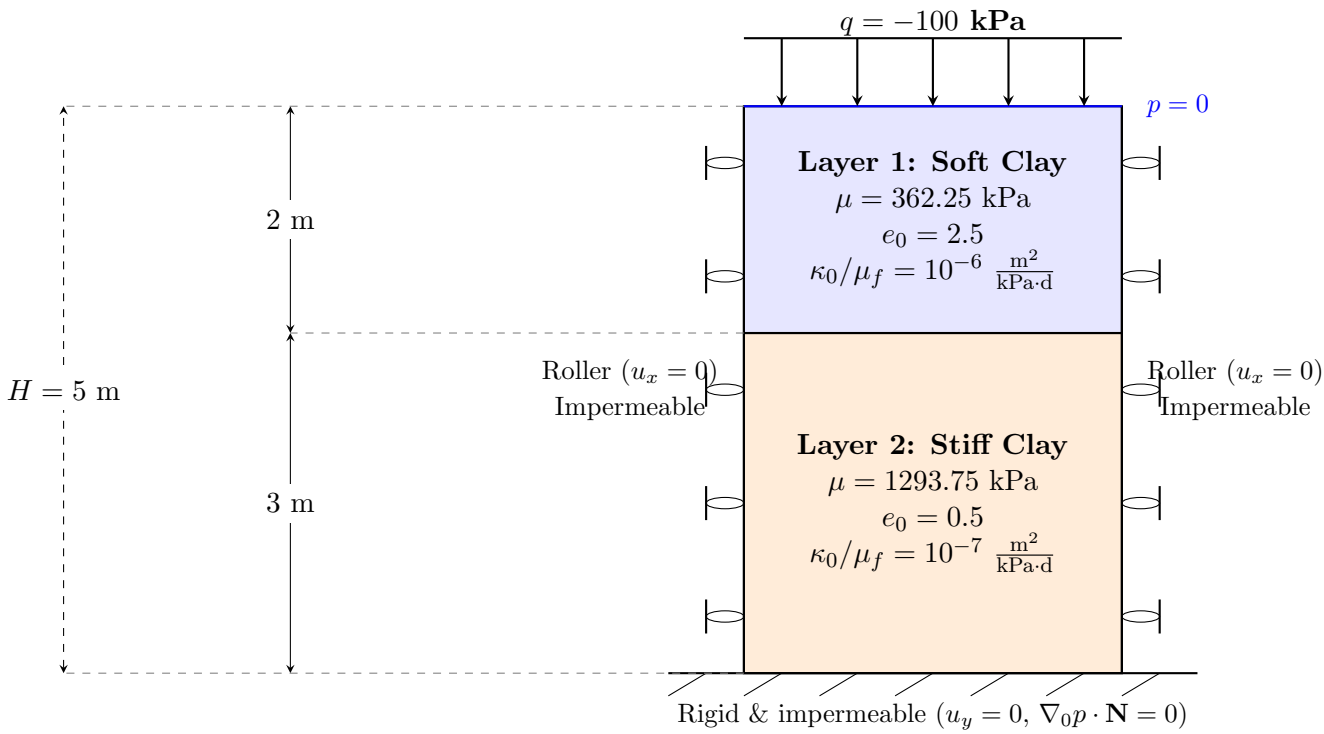

Figure 4: Schematic representation of the one-dimensional bilayer large-strain consolidation model.

Numerical strategy. The fully coupled nonlinear system is solved using PJFNK with MUMPS and automatic scaling. Implicit Euler time integration is used for stability. The time step starts at $\Delta t = 0.001$ day to resolve rapid pore-pressure generation during loading and then grows adaptively over a 365-day simulation.

9 Kernel–Variable Coupling Architectures

The figures in this section summarize how displacement and pressure variables interact in the numerical implementation. Solid arrows indicate the primary variable operated on by a kernel. Dashed arrows indicate coupled dependencies.

10 Summary of Key Differences

11 Results

This section collects the final numerical and analytical results in the same narrative order used in the presentation: first the benchmark analytical comparison, then the homogeneous classical consolidation cases, and finally the bilayer stratigraphy cases. The central purpose of these plots is to show where the small-strain formulation remains accurate and where finite-strain kinematics become necessary.

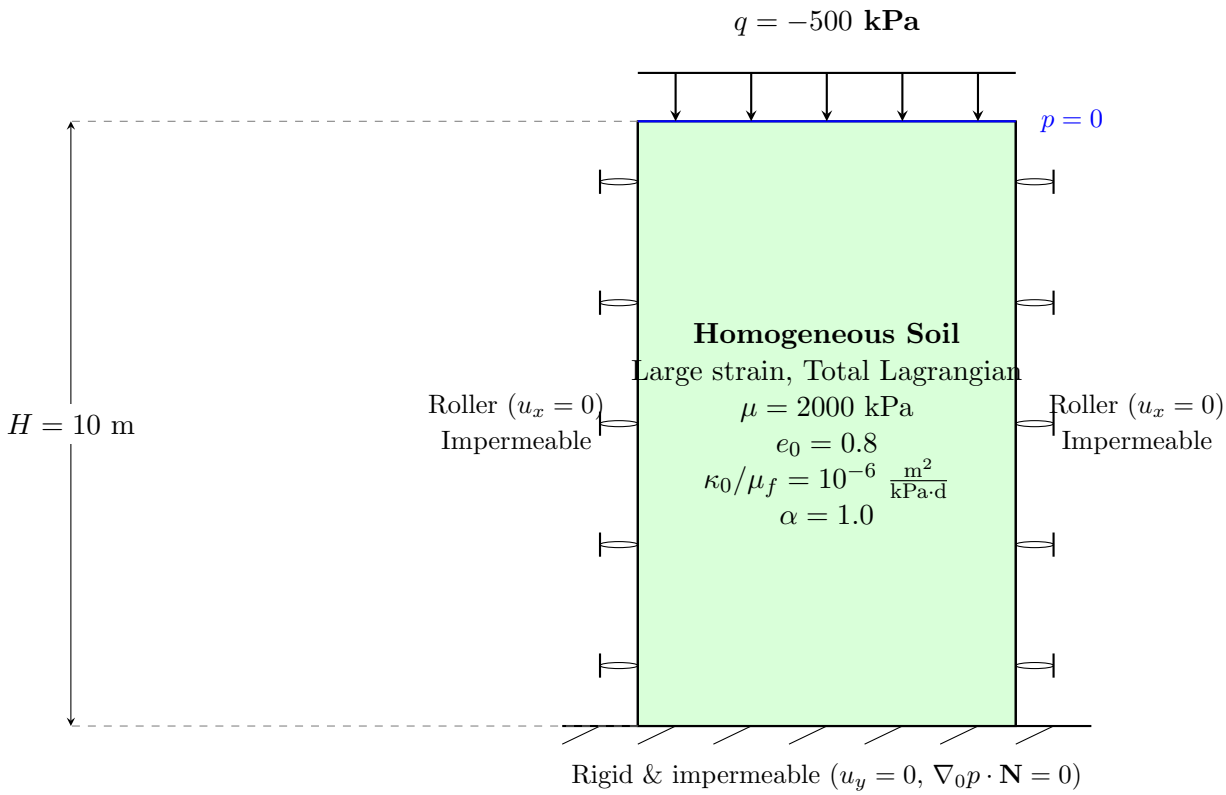

Figure 5: Large-strain homogeneous consolidation model.

Table 2: Comparison of small-strain and large-strain poromechanics formulations.

Quantity	Small strain	Large strain, Total Lagrangian
Domain	Current/reference effectively identical	Reference domain Ω_0
Kinematics	$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$	$\mathbf{F} = \mathbf{I} + \nabla_0\mathbf{u}$, $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T\mathbf{F} - \mathbf{I})$
Volume change	$\nabla \cdot \mathbf{u}$	$J = \det \mathbf{F}$
Stress measure	Cauchy stress $\boldsymbol{\sigma}$	PK1 stress $\mathbf{P}$ or PK2 stress $\mathbf{S}$
Effective stress coupling	$-\alpha p\mathbf{I}$	$-\alpha p J\mathbf{C}^{-1}$
Darcy tensor	$\mathbf{k}/\mu_f$	$\mathbf{K}_{\mathrm{TL}}/\mu_f = J\mathbf{F}^{-1}\mathbf{k}\mathbf{F}^{-T}/\mu_f$
Storage/coupling term	$\alpha\nabla \cdot \dot{\mathbf{u}}$	$\alpha\dot{J}$

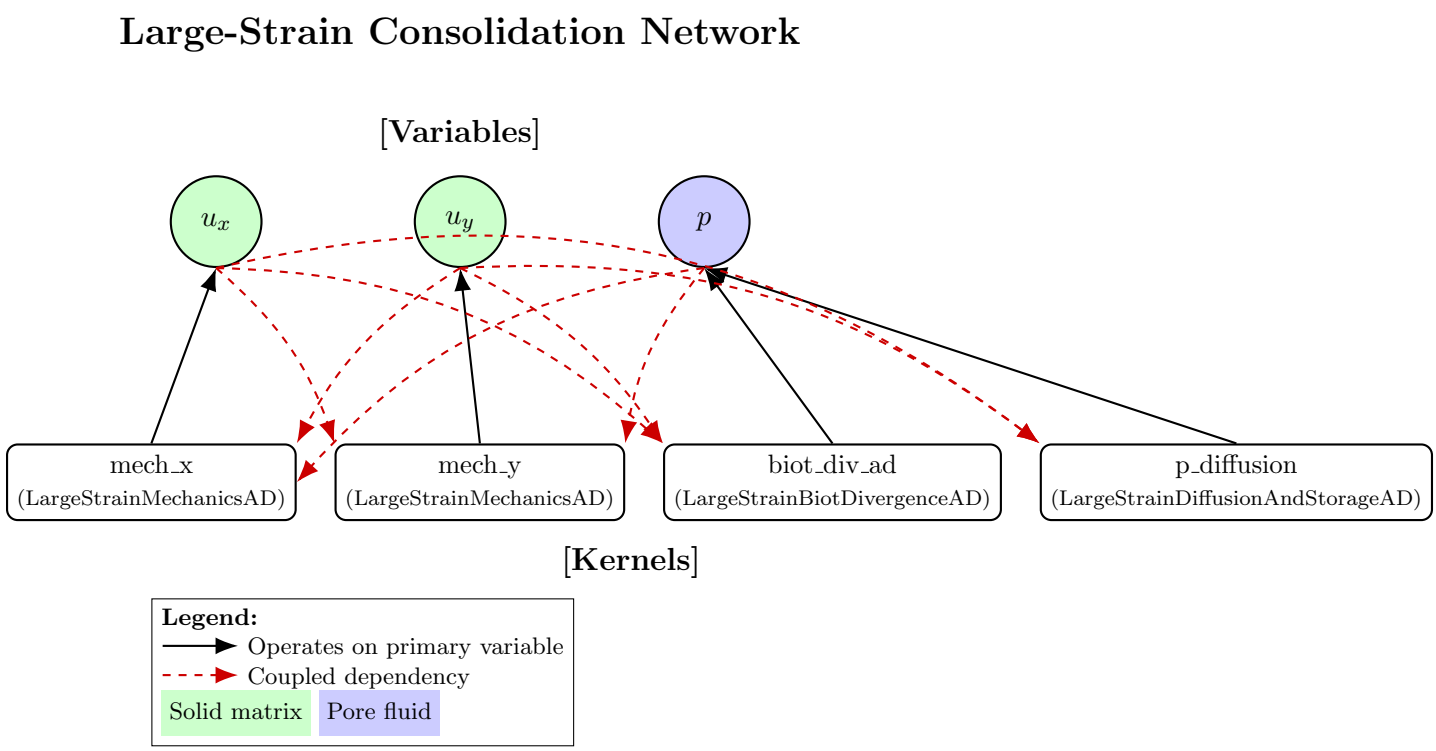

Figure 6: Interaction flowchart for the fully coupled large-strain Total Lagrangian model. The pressure-diffusion kernel depends on u_x and u_y because it must compute $\mathbf{F}$, J, and the pulled-back permeability tensor $\mathbf{K}_{\mathrm{TL}}$.

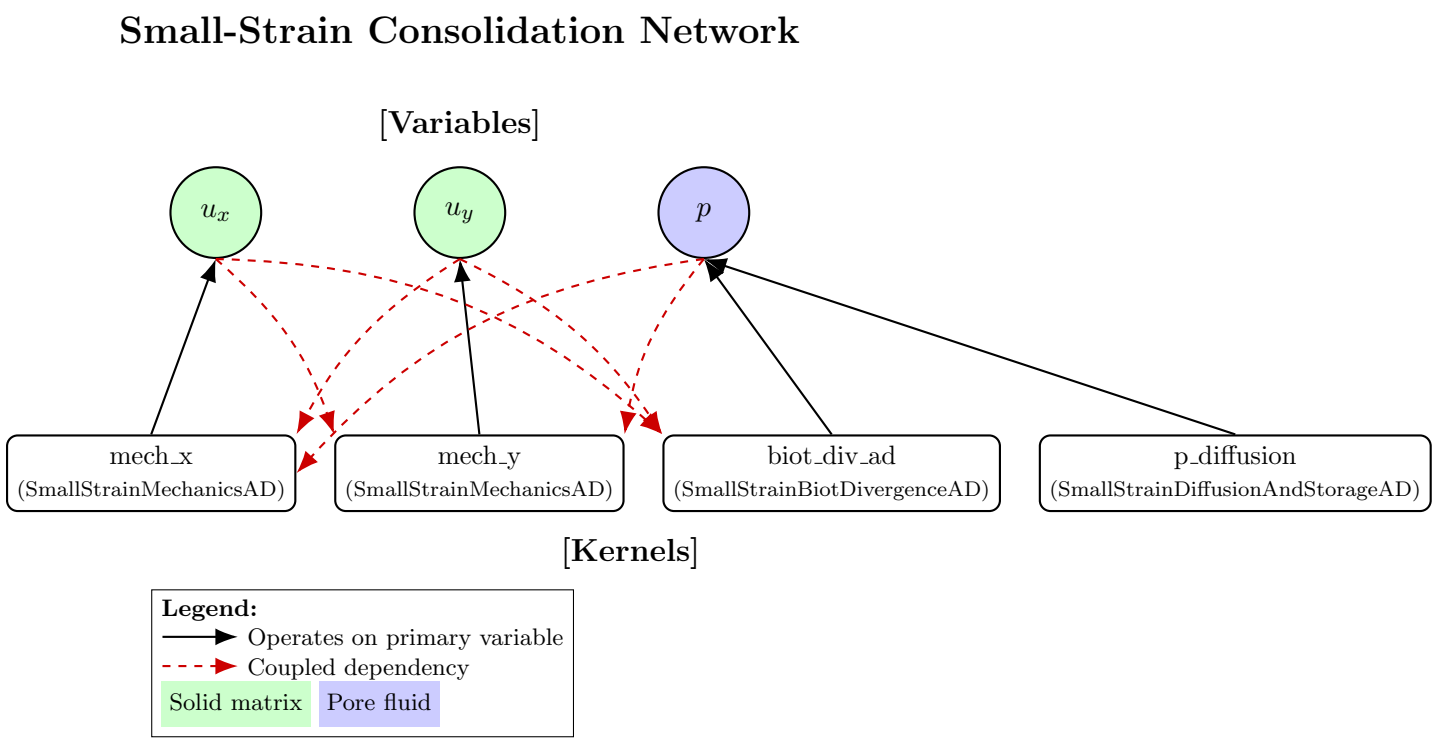

Figure 7: Interaction flowchart for the small-strain model. The pressure-diffusion kernel does not require displacement variables because permeability and geometry are assumed constant.

11.1 Analytical Benchmark: Degree of Consolidation

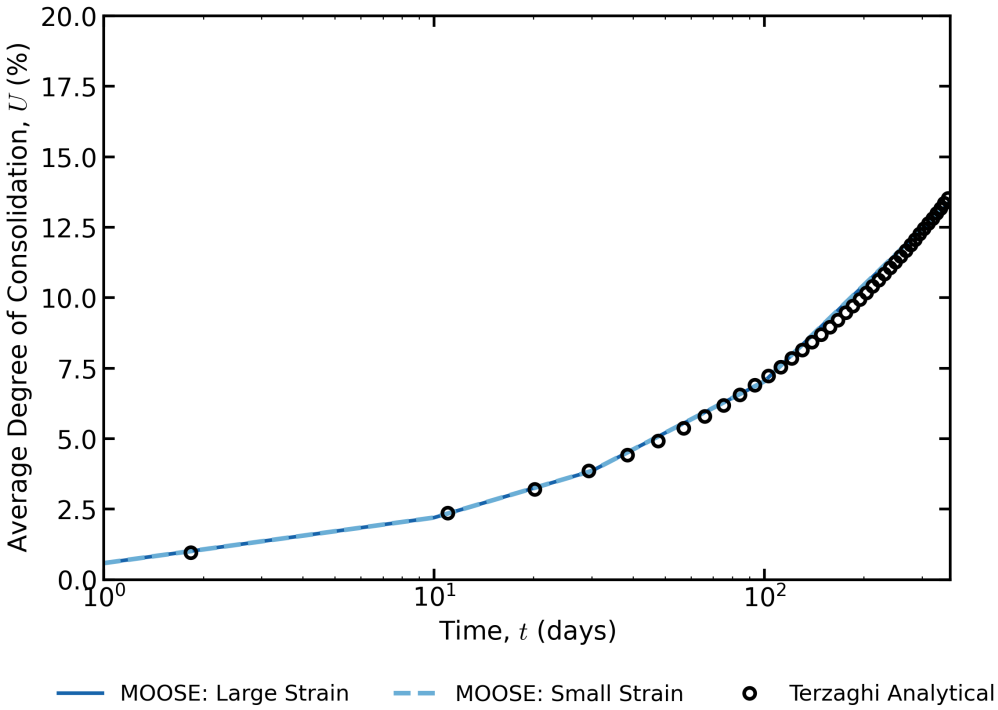

Figure 8: Analytical degree-of-consolidation benchmark for one-dimensional Terzaghi consolidation. This figure provides the reference trend for the average degree of consolidation U_{avg} as a function of time factor or time. It is used to verify that the numerical implementation reproduces the expected diffusion-controlled consolidation response before comparing small- and large-strain models.

The analytical solution represents the classical small-strain Terzaghi limit. In this limit, geometry, permeability, and drainage length are fixed, so consolidation is governed by linear pressure diffusion. Agreement with this curve demonstrates that the finite element implementation, boundary conditions, and time integration reproduce the expected one-dimensional response when the imposed load is small enough for infinitesimal strain assumptions to remain valid.

11.2 Analytical Benchmark: Strain Profile at Small Load

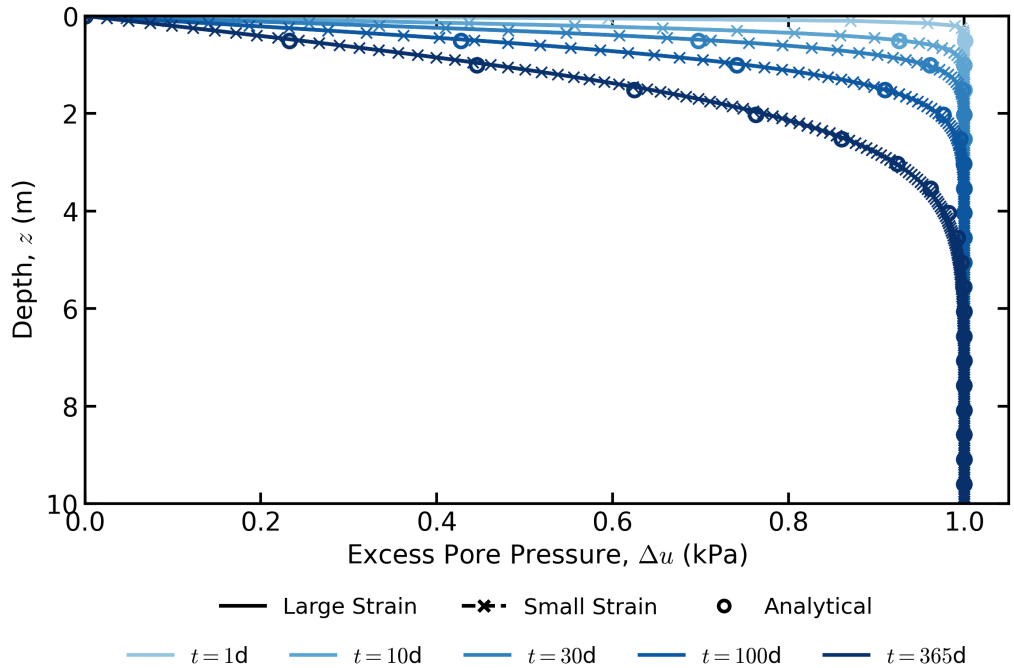

Figure 9: Comparison of strain profiles for the analytical small-load case, corresponding to a load magnitude of 1 kPa. At this load level, the deformation is small, and the finite-strain correction terms are negligible.

For the 1 kPa benchmark, the small-strain and large-strain strain measures are expected to be nearly indistinguishable. This result is important because it confirms that the finite-strain implementation reduces to the classical small-strain behavior in the appropriate limiting case. In other words, the large-strain formulation does not introduce artificial differences when the deformation is truly infinitesimal.

11.3 Homogeneous Classical Model: Load of $100\,\mathrm{kPa}$

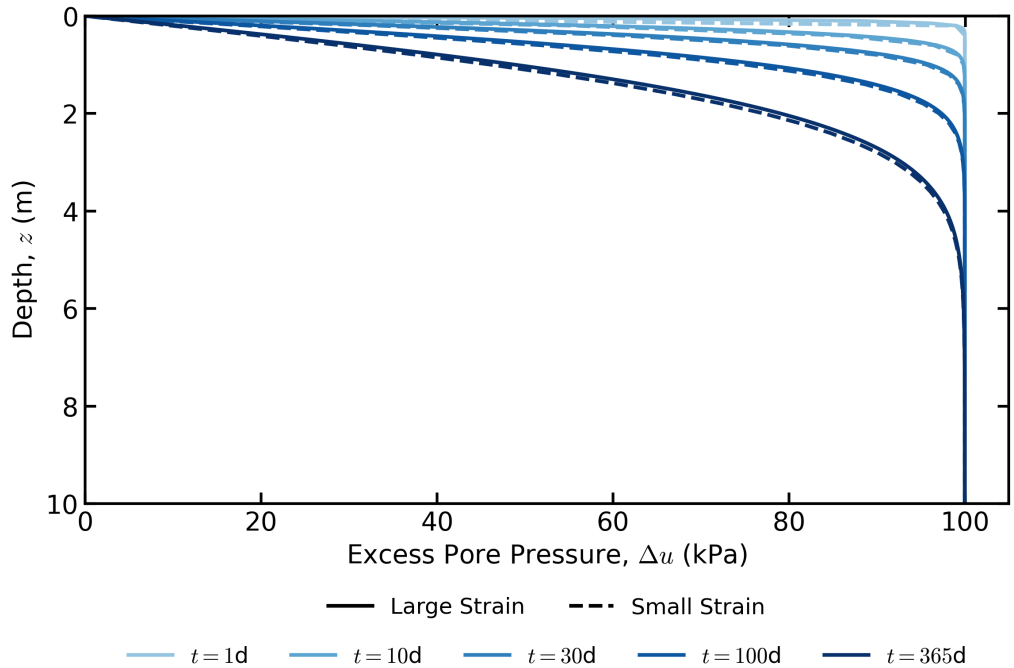

Figure 10: Large-strain versus small-strain comparison for the homogeneous classical consolidation model under a $100\,\mathrm{kPa}$ compressive surcharge. The pressure isochrones show the evolution of excess pore pressure with depth, while the consolidation curve summarizes the average consolidation response over time.

At $100\,\mathrm{kPa}$, the two formulations begin to separate. The small-strain model treats the geometry and hydraulic path length as unchanged, whereas the large-strain model updates the deformation through $\mathbf{F}$ and J. As a result, the large-strain solution reflects the changing volume of the porous skeleton and the deformation-dependent mapping of the hydraulic problem. The difference is not merely a numerical artifact; it is the expected consequence of retaining the nonlinear kinematic terms that are dropped in the infinitesimal strain approximation.

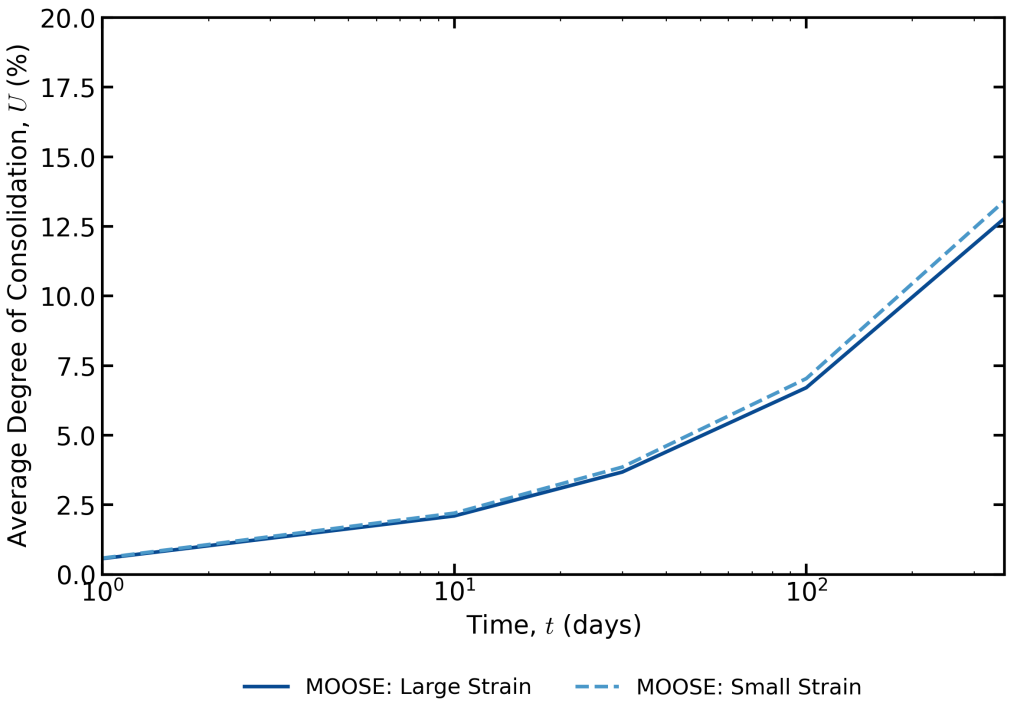

Figure 11: Average degree of consolidation for the homogeneous classical model under 100 kPa. The plotted curves compare the small-strain and large-strain predictions of the global consolidation rate.

The degree-of-consolidation comparison provides a global measure of the same behavior seen in the pressure profiles. Even when the pressure distributions appear qualitatively similar, the accumulated settlement and average dissipation can differ because the large-strain model accounts for finite changes in volume. This becomes increasingly important when interpreting consolidation time and final settlement under practical loading levels.

11.4 Homogeneous Classical Model: Load of $500\,\mathrm{kPa}$

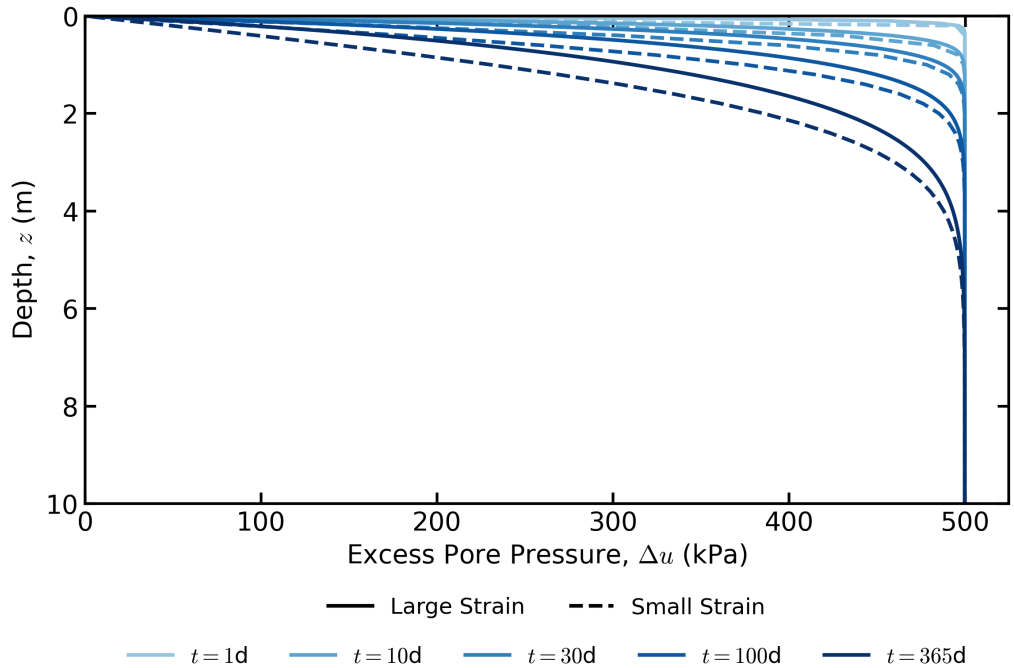

Figure 12: Large-strain versus small-strain comparison for the homogeneous classical consolidation model under a $500\,\mathrm{kPa}$ compressive surcharge. The high load magnifies the nonlinear kinematic effects.

Under the $500\,\mathrm{kPa}$ load, the divergence between small-strain and large-strain predictions becomes much more pronounced. The small-strain model continues to extrapolate from a fixed-geometry assumption, while the large-strain model allows settlement, volume change, and the deformation-dependent configuration to affect the governing equations. This is the regime where the finite-strain formulation is no longer an optional refinement; it directly changes the predicted consolidation response.

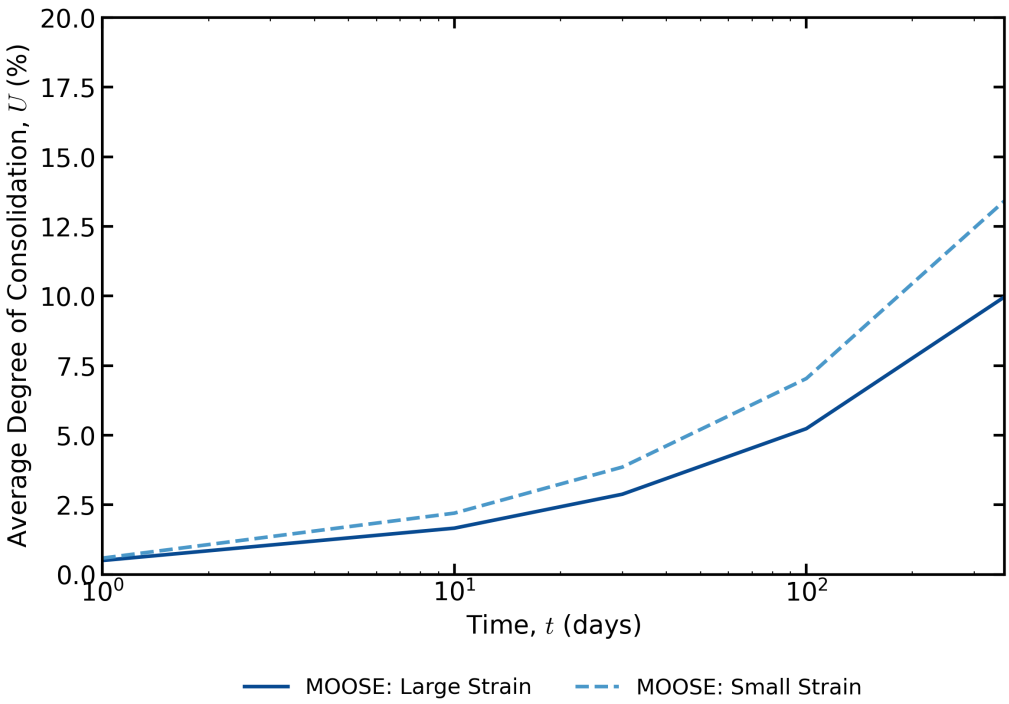

Figure 13: Average degree of consolidation for the homogeneous classical model under 500 kPa. The increasing separation between the curves illustrates the load-dependence of the small-strain error.

The 500 kPa degree-of-consolidation plot shows that the error introduced by the small-strain approximation is load dependent. For low loads, the quadratic terms in the Green–Lagrange strain are small. For high loads, those terms contribute significantly, so neglecting them leads to a different consolidation curve and a different prediction of field performance.

11.5 Bilayer Model: Load of 1 kPa

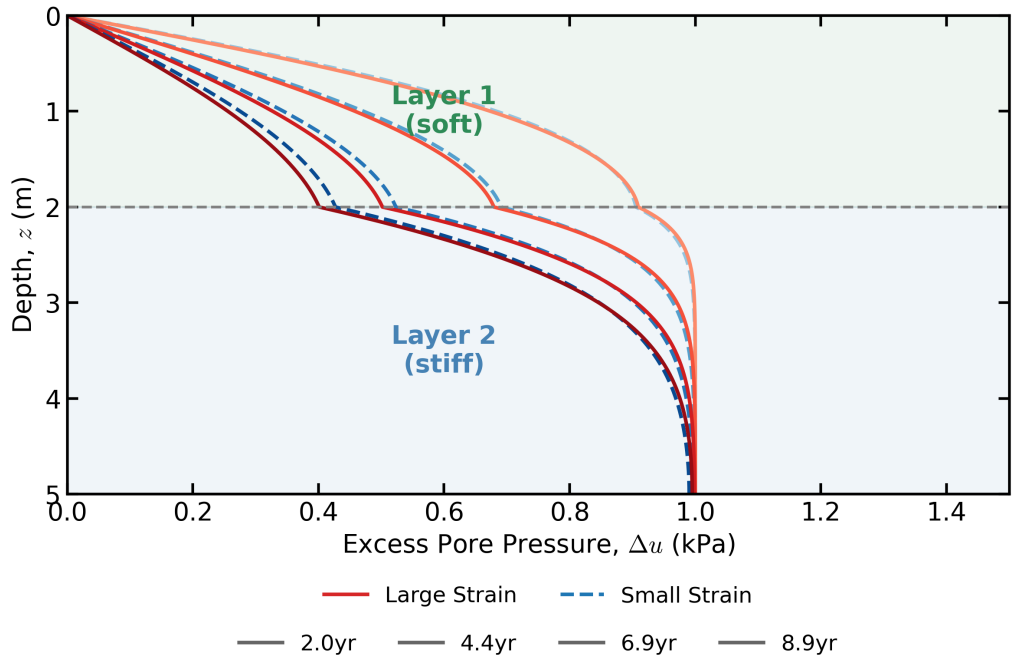

Figure 14: Excess pore-pressure and strain comparison for the bilayer consolidation model under a 1 kPa surcharge. The upper soft clay and lower stiff clay layers produce different local responses because of their contrasting stiffness and mobility.

The bilayer model introduces stratigraphic contrast. The upper soft clay has lower shear stiffness and higher mobility, while the lower stiff clay has higher shear stiffness and lower mobility. Even at 1 kPa, the layer interface is visible in the response because pore-pressure dissipation and strain localization depend on material properties. However, because the load is small, the difference between small-strain and large-strain kinematics remains limited.

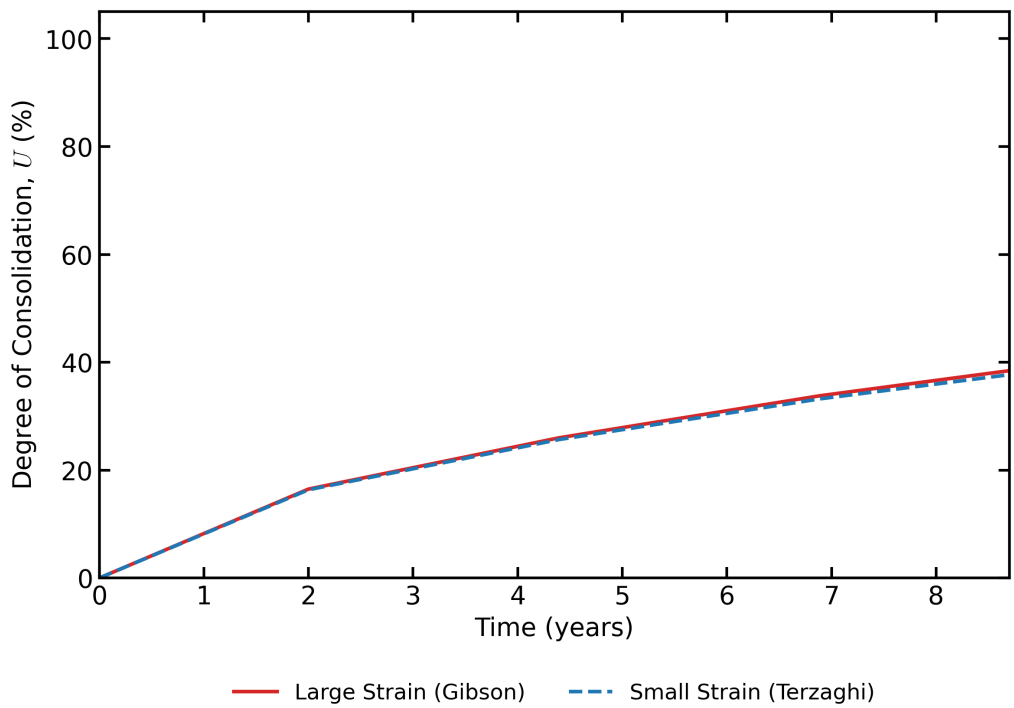

Figure 15: Average degree of consolidation for the bilayer model under 1 kPa. The close agreement between small- and large-strain predictions indicates that infinitesimal kinematics are adequate for very small loading.

This result confirms the same limiting behavior observed in the homogeneous benchmark: when deformations are small, both formulations give nearly the same global consolidation prediction. The bilayer case is nevertheless more complex because the drainage and deformation response are controlled by the contrast between the soft and stiff layers.

11.6 Bilayer Model: Load of $100\,\mathrm{kPa}$

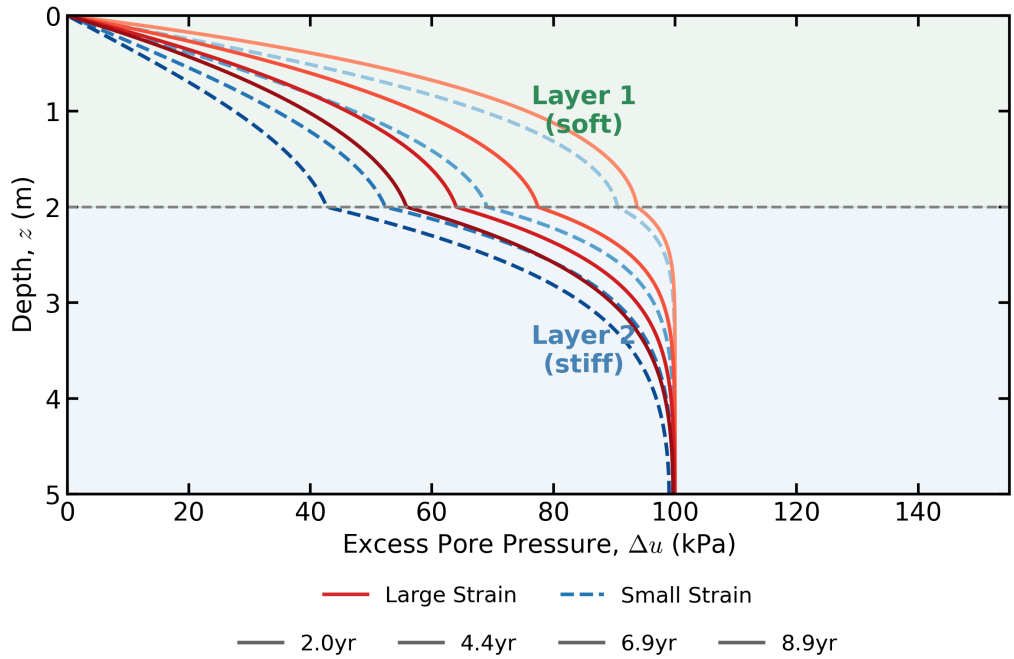

Figure 16: Excess pore-pressure and strain comparison for the bilayer consolidation model under a $100\,\mathrm{kPa}$ surcharge. The soft upper clay undergoes larger deformation than the stiff lower layer, making finite-strain effects more visible.

At $100\,\mathrm{kPa}$, the bilayer response shows a stronger separation between formulations. The soft clay layer is more compressible, so it experiences larger deformation and stronger geometric nonlinearity. The lower stiff layer deforms less, but its lower mobility slows pressure dissipation. The combined effect is a coupled mechanical–hydraulic response that cannot be represented accurately by a single homogeneous diffusion coefficient.

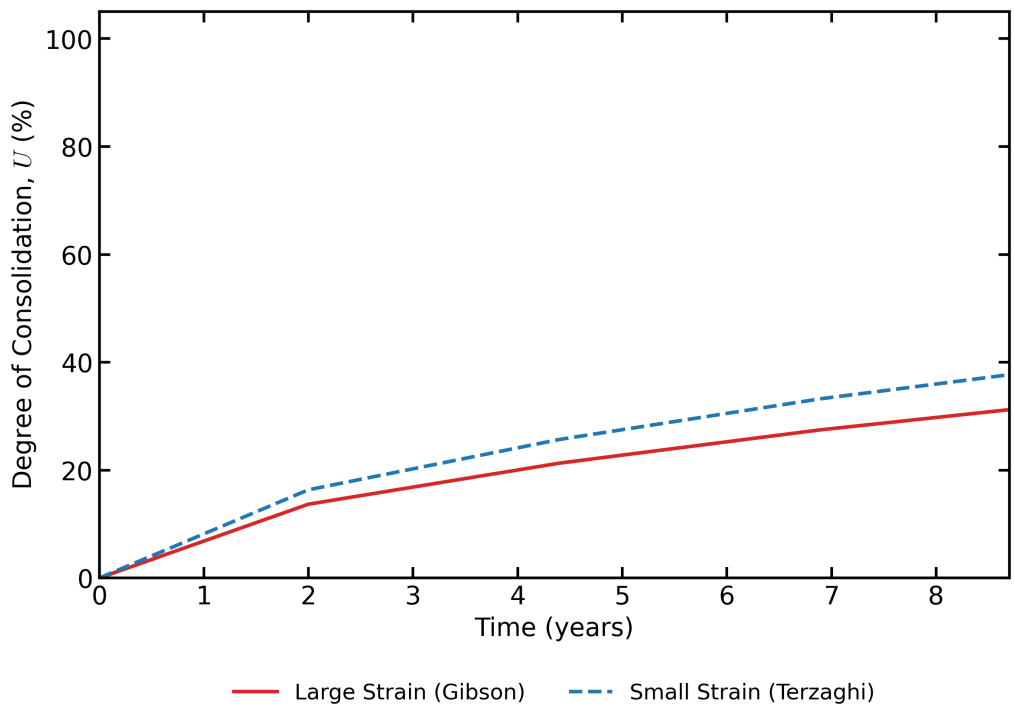

Figure 17: Average degree of consolidation for the bilayer model under 100 kPa. The separation between the curves shows that stratigraphy and finite deformation interact, especially when a highly compressible layer is present.

The bilayer degree-of-consolidation curve highlights the practical consequence of using small-strain theory in layered soils. In the soft layer, finite deformation alters the local void structure and volume change. In the stiff layer, low mobility controls the rate at which pressure can dissipate. The large-strain model captures both effects through the Jacobian, deformation gradient, and pulled-back permeability, whereas the small-strain model assumes a fixed domain and constant transport geometry.

12 Results Discussion

The results show that the difference between small-strain and large-strain consolidation is controlled primarily by load magnitude, material compressibility, and stratigraphic contrast. For the 1 kPa cases, the displacement gradients are small, so the quadratic terms in the Green–Lagrange strain are negligible. Therefore, small-strain Terzaghi theory and the large-strain formulation give nearly the same response.

As the applied load increases to 100 kPa and 500 kPa, the infinitesimal-strain assumptions become less appropriate. The small-strain model assumes that the domain geometry, drainage path, and hydraulic transport tensor remain fixed. The large-strain model instead tracks the motion from Ω_0 to Ω_t, computes $\mathbf{F}$ and J, and updates the weak form using reference-configuration quantities. This is why the two formulations diverge under larger loads.

The bilayer examples demonstrate that finite-strain effects are not only a function of the external load. They are also strongly influenced by material layering. The soft upper clay, with higher void ratio and lower stiffness, is more susceptible to finite deformation. The stiff lower clay, with lower

mobility, controls the rate of drainage. Together these properties produce a consolidation response that is both mechanically nonlinear and hydraulically heterogeneous.

From an engineering standpoint, the main message is that small-strain theory is an efficient and useful baseline for stiff soils, small loads, or early verification against analytical solutions. However, for soft soils, large settlements, heavy surcharges, swelling materials, or strongly layered deposits, large-strain theory is required to avoid misleading predictions of settlement magnitude and consolidation rate.

13 Conclusion

This document compared small-strain and large-strain formulations for one-dimensional poromechanical consolidation. The analytical Terzaghi solution was used as a verification benchmark, and the numerical models were then extended to homogeneous and bilayer systems under different load levels.

The main conclusions are:

1. For very small loads, the finite-strain formulation correctly reduces to the small-strain response, giving excellent agreement with the analytical Terzaghi benchmark.

2. The error associated with small-strain kinematics increases with load because the omitted quadratic strain terms become important.

3. In layered soils, material contrast amplifies the importance of the formulation: the soft layer controls deformation, while the lower-permeability stiff layer controls pressure dissipation.

4. The deformation gradient $\mathbf{F}$ and Jacobian J are essential in the large-strain weak form because they connect the reference and current configurations and account for finite changes in volume and transport geometry.

5. Small-strain Terzaghi theory remains valuable for verification and for problems with negligible deformation, but large-strain consolidation theory is necessary when settlement is large enough to change the geometry of the problem.

Overall, the results support the engineering conclusion that finite-strain poromechanics is not merely a mathematical refinement. It is required for reliable prediction when soft soils, high loads, or strong stratigraphic contrasts produce significant deformation.

14 References

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